

Orientation as Holonomy of the Reader Configuration Space

Driven by Dean A. Kulik

September 2026

Abstract

A finite reader partitions a cell set into m sectors and observes a permutation T through the sector-to-sector transition table $C(T)$. Because T is a bijection, the antisymmetric part $A = C - C^T$ has zero row sums and therefore defines a conserved directed-circulation channel. At three sectors that channel is one-dimensional and transforms by the sign representation of S_3 , so its sign is a natural orientation observable.

We show that this sign is not a function of T . Holding one permutation and one unordered three-sector partition fixed, the six labelings of that partition give the same magnitude and both signs. No refinement of T can supply the sign, because a function cannot take two values at one argument.

The missing structure is the reader's own space of admissible configurations. For an ambient adjacency graph X , let $\mathcal{C}(X, m)$ have as vertices the partitions of the cells into m connected nonempty regions, with edges given by an admissible move rule. Labelings form an S_m -cover, and transport around closed walks defines a monodromy

$$\rho : \pi_1(\mathcal{C}(X, m)) \rightarrow S_m,$$

of which the orientation observable is the parity projection $\text{sgn} \circ \rho$. The image of ρ depends on X and the move rule, never on T .

We compute it. Under either of two natural move rules, unrestricted subsets and connected regions of a two-dimensional torus give S_3 ; requiring three sectors to remain contiguous arcs of an oriented cycle gives A_3 ; the three-dimensional cube Q_3 has full S_3 monodromy exhaustively; and the larger three-dimensional torus Z_3^3 admits an explicit odd exchange loop, so its image is not contained in A_3 . Restricting moves to adjacent cells does not by itself remove odd holonomy — what removes it is the admissibility condition on the partitions, not the locality of the move.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Separately, a finite translation system carries a full three-sector channel while its state-transition graph is triangle-free, showing that the complete graph supporting the channel is a graph on sector labels and is independent of state adjacency.

1. Introduction

A reader observes a transformation through a restricted interface. In the permutation setting the transformation is exact and finite, the reader is a partition, and the observation is a table of sector-to-sector counts. This makes the inverse question sharp: which features of the observation belong to the transformation, and which belong to the way the reader is arranged?

Prior partition-reader work gives an exact accounting for coarse observations, with what the reader resolves and what remains in the fiber summing to $\log_2 N!$ for every reading [1]. Under all balanced two-block cuts the fiber is precisely the independent cycle reversals, so the unoriented cycle structure survives and orientation does not [2]. That residual motivates a finer reader.

Three sectors is the first resolution at which a conserved directed circulation can exist at all, and at three sectors that circulation is a single scalar. Its sign looks like the orientation the two-sector reader could not see. This paper shows the sign is not carried by the transformation, locates what does carry it, and computes that object across several ambient geometries.

Relation to known work. The behaviour we find in one versus two dimensions is not itself new. That indistinguishable objects cannot be exchanged on a line and can be exchanged in the plane, with the braid group governing the difference, is classical [5,6]. Configuration spaces of points on graphs and surfaces are well developed [5,6,7]; so is the topology of the partition lattice as a poset [8]; and the representation appearing in §3 is a standard Specht module [9]. The present work treats partitions into connected regions, under an explicit move rule, as a configuration space in its own right, computes its monodromy, and identifies a reader's observable with that monodromy. That identification is the contribution. We make no priority claim against the literature cited above; the result stands on the constructions and computations given here.

2. Definitions

2.1 Ambient adjacency and readers

Let $X = (\Omega, E_X)$ be a finite graph. Its vertices are *cells*. A subset is *connected* if it induces a connected subgraph of X . Taking X complete recovers the unconstrained case, in which every nonempty subset is connected.

An m -sector reader is an ordered tuple $R = (R_1, \dots, R_m)$ of pairwise disjoint nonempty connected subsets covering Ω . Forgetting the order gives the underlying partition.

2.2 Two move rules

Two natural notions of an elementary reader move appear in what follows, and they generate different graphs on the same vertex set. We name both and label every result with the one used.

Transfer τ . Move a single cell from one sector to another. Sector sizes change. Admissible when every affected sector remains nonempty and connected.

Swap σ . Exchange one cell of one sector with one cell of another. Sector sizes are preserved. Admissible under the same conditions.

A swap is not a τ -edge; it is a pair of transfers whose intermediate state may itself be inadmissible, so σ is genuinely a different move rule rather than a shorthand. Where sector sizes are constrained to be equal, only σ is available. Where they are free, τ is the finer rule. §7 reports the same answer under both in the case where both apply, but this is a computed fact, not an identity.

2.3 Configuration space, cover, monodromy

Fix a move rule $\mu \in \{\tau, \sigma\}$. The configuration space $\mathcal{C}(X, m; \mu)$ is the graph whose vertices are the admissible underlying partitions and whose edges are the μ -moves. The ordered space $\mathcal{C}$ has labeled tuples as vertices, with a move carrying labels along; forgetting labels gives an $m!$ -fold covering of graphs on which S_m acts freely by relabeling.

A closed walk in $\mathcal{C}$ lifts to a walk in $\mathcal{C}$ whose endpoint differs from its start by a relabeling, giving

$$\rho : \pi_1(\mathcal{C}, p) \rightarrow S_m,$$

well defined up to conjugacy by the choice of base labeling. Operationally, $\text{im } \rho$ is trivial exactly when $\mathcal{C}$ has $m!$ times as many connected components as $\mathcal{C}$. No observer enters this definition: ρ is a property of the configuration space and its cover, present whether or not anything occupies an address in it.

2.4 Two complete graphs

Two complete graphs occur in this paper and must be kept apart. The *state-transition graph* Γ has states as vertices and generator actions as edges. The graph K_m of §3 has sector labels as vertices and is complete by construction, since C_{ij} is defined for every ordered pair of sectors. §10 shows the two are independent.

3. The conserved antisymmetric channel

For an ordered reader and a permutation T of Ω , set $C_{ij}(T) = \#\{x \in R_i : T(x) \in R_j\}$. Bijectivity gives $\sum_j C_{ij} = |R_i|$ and $\sum_i C_{ij} = |R_j|$, so row and column sums agree. Hence $A = C - C^T$ is antisymmetric with every row sum zero. Write $\mathcal{A}_m$ for that space.

Proposition 3.1. $\dim \mathcal{A}_m = C(m, 2) - (m-1) = (m-1)(m-2)/2$, and $\mathcal{A}_m$ is canonically the cycle space $Z_1(K_m; \mathbb{R})$.

Proof. An antisymmetric matrix is determined by its $C(m, 2)$ entries above the diagonal. The zero-row-sum equations impose m linear conditions whose sum vanishes identically by antisymmetry, hence $m-1$ independent ones. Identifying A_{ij} with a flow on the edge ij of K_m , zero row sum is zero divergence at each vertex, and the kernel of the divergence operator on a connected graph is its cycle space, of dimension $|E| - |V| + 1 = C(m, 2) - m + 1$. ■

Two consequences. For $m = 2$, K_2 is a tree and the channel vanishes identically: a balanced two-sector reader has no conserved directed circulation, which is the algebraic content of the orientation-blindness reported in [2]. For $m = 3$, the dimension is one and every element is

$$A(a) = \begin{bmatrix} 0 & a & -a \\ -a & 0 & a \\ a & -a & 0 \end{bmatrix}.$$

Proposition 3.2. For $m \geq 3$, $\mathcal{A}_m$ is an irreducible S_m -module, isomorphic to the Specht module $S^\lambda(m-2, 1, 1)$. It contains the sign representation if and only if $m = 3$, where $\sigma \cdot A(a) = A(\text{sgn}(\sigma) \cdot a)$.

The identification is classical [9] and we use it as background rather than claim it. We verified irreducibility and the sign-isotypic dimension independently for $m \leq 6$ by isotypic projection and by character inner product, obtaining agreement.

The representation theory settles what the channel *is*. It does not settle where the sign comes from, and §4 shows the natural answer is wrong.

4. The sign is not a function of the transformation

Throughout this section X is complete, so every subset is admissible. Take $\Omega = \{0, \dots, 5\}$, $T = (1, 2, 0, 4, 5, 3)$ in one-line notation, and the unordered partition $\{0, 3\} \cup \{1, 4\} \cup \{2, 5\}$. Evaluating $a = A_{12}$ for each of the six labelings:

labeling	a	$ a $	$\text{sgn}(a)$
$(0, 1, 2)$	2	2	+1
$(0, 2, 1)$	-2	2	-1
$(1, 0, 2)$	-2	2	-1
$(1, 2, 0)$	2	2	+1
$(2, 0, 1)$	2	2	+1
$(2, 1, 0)$	-2	2	-1

Proposition 4.1. $\text{sgn}(a)$ is not a function of T .

Proof. The permutation is the same in all six rows, and so is the unordered partition. Only the labeling differs, and both signs occur. ■

This is stronger than the statement that some particular reader fails to recover orientation. It says no refinement of T can determine the sign, since a function cannot take two values at one argument. Writing ℓ for a labeling and $\gamma\ell$ for any path in $\mathcal{C}$ from the base labeling to ℓ , the observation decomposes as

$$a(T, \ell) = |a(T)| \cdot \text{sgn}(\rho(\gamma\ell)).$$

The first factor is an invariant of T ; the second is fixed by the reader's address in the cover. The reader manufactures nothing — it does not alter $\mathcal{C}(T)$ — but which lift of the channel is present at its address is determined by where it stands. The question becomes whether a labeling can be chosen consistently across the whole configuration space, and that is a question about ρ .

5. The exchange construction

Theorem 5.1. Let B and C be sectors with $|B| = |C| = k$. Suppose the k swaps below are each admissible. Then they form a closed walk in $\mathcal{C}(X, m; \sigma)$ whose monodromy is the transposition $(B\ C)$, an odd permutation.

Proof. Enumerate $B_0 = \{b_1, \dots, b_k\}$ and $C_0 = \{c_1, \dots, c_k\}$. For $i = 1, \dots, k$ apply the swap exchanging b_i with c_i . By induction, after step i the sector labeled B holds $\{c_1, \dots, c_i\} \cup \{b_{i+1}, \dots, b_k\}$ and the sector labeled C holds the complement of that set within $B_0 \cup C_0$. At $i = k$ the sector labeled B holds C_0 and the sector labeled C holds B_0 . No other sector was touched, so the underlying partition is the one we began with and the induced relabeling is exactly $(B\ C)$. ■

When X is complete every intermediate is admissible and the construction is unconditional; we verified it for $k = 2, 3, 4, 5, 8$. Under a connectivity constraint the only question is whether the intermediates stay connected, so Theorem 5.1 reduces the appearance of odd holonomy to a purely geometric admissibility question. That reduction is what §§6–9 answer.

Corollary 5.2. A closed walk can return the reader to the same *unlabeled* configuration while failing to return it to the same *labeled* one. Hence the underlying partition may be globally well defined while no globally consistent sector labeling exists.

6. One dimension: contiguity restricts the image

Let X be the oriented cycle C_N and require each of three sectors to remain a contiguous arc. Moves are boundary slides, i.e. τ -transfers between neighbouring arcs. Exhaustively for $N = 6$:

$$V = 20, \quad E = 36, \quad \beta_1 = E - V + 1 = 17.$$

The ordered cover has exactly twice as many components as the base, so $\text{im } \rho = A_3$. A spanning-tree census over all 17 fundamental cycles returns only even elements. The same holds at $N = 9$ and $N = 12$.

Locality is not the mechanism. It is tempting to attribute this to the moves being local. It is not that. Allowing *unrestricted* subsets on the same cycle, with swaps restricted to cyclically adjacent cells, gives $\text{im } \rho = S_3$ at $N = 6$. The odd loops survive an entirely local move rule. What removes them is the admissibility condition — that sectors be arcs — not the locality of the move.

The mechanism is that three arcs on an oriented circle inherit a cyclic order from the ambient orientation, and no sequence of boundary slides can permute them oddly while all three remain arcs. An odd relabeling would reverse the inherited cyclic order, which no admissible path realizes.

An explicit generator. Component counts establish $\text{im } \rho = A_3$ without exhibiting an element. Writing a configuration as the word of sector labels on cells $0, \dots, 5$, the following six transfers each move one cell, keep all three sectors nonempty contiguous arcs, and return the starting partition:

$$001122 \rightarrow 201122 \rightarrow 200122 \rightarrow 220122 \rightarrow 220112 \rightarrow 220012 \rightarrow 220011.$$

The final word is the initial partition with labels cycled, so the monodromy is the 3-cycle $(0 \ 1 \ 2)$ — even, as required. The tempting two-transfer shortcut, relabeling one whole arc to match another, is inadmissible: from 001122 it reaches 002222 , which empties one sector and leaves another disconnected. Odd monodromy is blocked by admissibility at every intermediate step, not merely absent from the endpoints.

7. Two dimensions: odd exchange becomes available

Let $X = Z_3 \times Z_3$ with periodic adjacency, three connected regions, no arc condition. Under τ -transfers with sizes free:

$$V = 986, \quad E = 5121, \quad \beta_1 = 4136,$$

and the ordered cover is connected, so $\text{im } \rho = S_3$. Under σ -swaps with the three regions constrained to equal size 3, the ordered cover is again connected and $\text{im } \rho = S_3$. The two move rules therefore agree here; as noted in §2.2 this is a computed agreement, not an identity.

A spanning-tree census over the 4136 fundamental cycles of the τ -space generates S_3 , and each of the three transpositions occurs as the monodromy of a *single* fundamental cycle. The odd exchange is therefore elementary in this space, not something assembled from several loops.

Basis dependence. The per-class counts are not invariants. Two spanning trees on the same space gave 1211 and 1216 odd cycles out of 4136, with different distributions across conjugacy classes. Only the generated subgroup, and whether any odd element occurs, are properties of the space; the census is a computational basis, not a measurement.

8. Three dimensions: a constructive witness

An exhaustive three-dimensional case. Take $X = Z_2^3$. Here the periodic neighbours ± 1 coincide in each coordinate, so the ambient is the 3-cube Q_3 , 3-regular on 8 cells. Exhaustive enumeration under τ -transfers gives 268 admissible unordered configurations and 1608 labeled ones, a ratio of exactly 6. Both the base and the ordered cover are connected, hence

$$\text{im } \rho = S_3 \text{ for } X = Q_3.$$

This is a full determination in three dimensions, not a witness, and it is the smallest such case we found. The larger Z_3^3 is beyond exhaustive enumeration, and there we give a construction instead.

For $X = Z_3^3$ with three connected slabs of nine cells, the construction of Theorem 5.1 completes in nine admissible swaps, returning the same unordered partition with an odd relabeling. Hence

$$\text{im } \rho \not\subseteq A_3.$$

This is stated at exactly that strength. An exhaustive census at 27 cells is not currently feasible, and an explicit loop establishes membership of one element without determining the image.

8.1 Strength of evidence

Three levels of evidence appear in this paper and should not be conflated. In particular, a search that finds a loop is a proof, while a search that fails to find one is not; the A_3 results therefore rest on exhaustive component counts, never on a failed search.

claim	evidence required	used in
$\text{im } \rho \not\subseteq A_m$	one explicit admissible odd loop	§8
$\text{im } \rho = S_m$	component count of the cover, or a generating census	§6, §7
$\text{im } \rho \subseteq A_m$	component count, or a proof that every admissible loop is even	§6

9. Classification

Collecting the computations, with the move rule stated for each row:

ambient X	admissible partitions	move	im ρ	method
complete	arbitrary subsets, $ R_i = 2$	σ	S_3	exhaustive, $N = 6$

cycle C_6	arbitrary subsets, $ R_i = 2$	σ , adjacent cells only	S_3	exhaustive, $N = 6$
oriented cycle C_N	contiguous arcs	τ (boundary slide)	A_3	exhaustive, $N = 6, 9, 12$
torus $Z_3 \times Z_3$	connected, sizes free	τ	S_3	exhaustive, 986 vertices
torus $Z_3 \times Z_3$	connected, sizes equal	σ	S_3	exhaustive
3-cube $Q_3 = Z_2^3$	connected, sizes free	τ	S_3	exhaustive, 268 vertices
torus Z_3^3	connected, sizes equal	σ	$\not\subseteq A_3$	explicit 9-swap loop

The variable that separates the rows is not the dimension of X as such, and not the locality of the move, but the topology of the space of admissible configurations that the pair (ambient, admissibility) produces.

10. Sector triangles are not state triangles

The K_m of Proposition 3.1 has sector labels as vertices and is complete by construction. It says nothing about adjacency among cells or states. The following separates the two.

Proposition 10.1. There is a finite state space with commuting bijective generators whose transition graph is triangle-free and which carries a three-sector reader with nonzero conserved channel.

Proof. Take $Z_4 \times Z_4$ with $T_a(x,y) = (x+1,y)$ and $T_b(x,y) = (x,y+1)$. These commute and are bijections. The transition graph has 16 vertices, 32 edges, cycle rank 17, and no triangle. Partition the cells by $(i+j) \bmod 3$, giving sectors of sizes 6, 5, 5, and read T_a :

$$A_{12} = 4, \quad A_{13} = -4, \quad A_{23} = 4. \blacksquare$$

The channel is a full circulation on a triangle-free state graph. Nor is the partition special: over 2000 random three-sector partitions of the same state space, 1283 give a nonzero channel. Any argument passing from the existence of the channel to a triangle in Γ has conflated two complete graphs on different vertex sets.

11. What the reader contributes

The results give a clean division of labour, which is worth stating explicitly because it is easy to misread in either direction.

object	role
T	the permutation; fixes the transition counts at any given configuration
unordered partition	the reader's configuration, without labels
ordered labeling	a lift of that configuration into the S_m -cover
closed walk in $\mathcal{C}$	transport through admissible configurations
ρ	the relabeling produced by that transport
$\text{sgn} \circ \rho$	the orientation observable at $m = 3$

The reader does not create a quantity. It does not perturb T , and it does not inject the sign. Its configuration determines which lift is present at its address, and a closed walk can change that lift while the underlying unlabeled

configuration returns to itself. Orientation is not the primary structure here; it is the parity shadow of a permutation-valued transport:

$$\pi_1(\mathcal{C}) \rightarrow S_3 \rightarrow \{\pm 1\}.$$

12. Method and reproducibility

All computations are finite and structural. The workflow is: enumerate admissible unordered partitions; build the move graph under the stated rule; build the ordered cover by applying every relabeling; compare component counts of base and cover; where a census is wanted, choose a spanning tree, lift each non-tree edge, record the induced relabeling, and generate the subgroup. The component-count method is preferred because it requires no choice of base path and no choice of tree.

A companion notebook reproduces every number here using only the Python standard library, with fixed seeds where sampling is used. Sampling appears only in the 2000-partition figure of §10 and is never substituted for an exhaustive claim.

result	status
$\dim \mathcal{A}_m$ and the cycle-space identification	proof, verified $m \leq 8$
Specht identification, sign rep at $m = 3$	classical; independently checked $m \leq 6$
six-labeling counterexample	direct computation
exchange construction	proof, subject to admissibility; checked $k \leq 8$
contiguous arcs $\rightarrow A_3$	exhaustive, $N = 6, 9, 12$
3×3 torus $\rightarrow S_3$	exhaustive, both move rules
3-cube $Q_3 \rightarrow S_3$	exhaustive, 268 configurations
$3 \times 3 \times 3$ torus $\not\subseteq A_3$	explicit witness loop
triangle-free channel	direct computation + 2000 random partitions

13. What is and is not established

Established. The channel dimension and its identification with the cycle space of K_m ; that at $m = 3$ it is one-dimensional and transforms by the sign character; that $\text{sgn}(a)$ is not a function of T ; the exchange construction; the computed images of ρ in the cases tabulated in §9; and the independence of the sector K_3 from any triangle in the state graph.

Not established. That any principle selects one ambient geometry over another; any general relationship between $\text{im } \rho$ and orientability of the ambient — the 2-torus is an orientable surface whose reader configuration space nevertheless has $\text{im } \rho = S_3$, so the two notions are distinct; that three-sector readers are canonical or physically fundamental; that $\text{im } \rho = S_3$ for Z_3^3 — the 3-cube Q_3 is determined exhaustively in §8, the larger torus only by witness; or that any of this constitutes a new development in braid-group theory. The contribution is the identification of the reader's observable with a configuration-space monodromy, and the computation of that monodromy in specific cases.

14. Open problems

- (i) Determine $\text{im } \rho$ exhaustively for Z_3^3 . §8 settles the 3-cube Q_3 exhaustively and Z_3^3 only by witness, so what remains open is the larger three-dimensional ambient, not three dimensions as such.
- (ii) Characterize the admissibility conditions under which $\text{im } \rho \subseteq A_m$. Contiguity on an oriented cycle is sufficient; no necessary condition is known.
- (iii) For $m \geq 4$ the channel is irreducible of dimension greater than one and has no one-dimensional subrepresentation, so there is no single label-free orientation bit. What observable replaces $\text{sgn} \circ \rho$ there is open.
- (iv) Determine whether any general relationship holds between reader monodromy and ambient orientability.
- (v) Characterize the minimal geometric conditions permitting an odd exchange of connected regions.

15. Conclusion

At three sectors the conserved antisymmetric channel is a single scalar, and its magnitude is an invariant of the transformation. Its sign is not. Holding the transformation and the unordered partition fixed and varying only the labeling produces both signs, so the sign cannot be a function of the transformation at any resolution.

What supplies it is the reader's own configuration space. Treating admissible configurations as a graph and their labelings as a covering space, closed walks carry a permutation-valued monodromy, and the orientation observable is its parity. The image of that monodromy is a property of the ambient adjacency and the admissibility rule alone. Contiguous arcs on an oriented cycle give A_3 ; connected regions on a two-dimensional torus give S_3 ; and in three dimensions the cube Q_3 has full S_3 monodromy by exhaustive enumeration, while the larger torus Z_3^3 already admits an explicit odd exchange loop.

The picture is therefore not that orientation was hidden in the permutation and eventually recovered by a sufficiently fine reader. The transformation supplies what its transition structure supplies. The reader's admissible configuration determines the lift through which that structure is observed, and whether such a lift can be chosen consistently at all.

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